

Graph Representation Learning for Student Sectioning: A Node2Vec-Based Approach

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Abstract

Student sectioning partitions enrolled students into instructional sections while preserving scheduling flexibility for subsequent course timetabling. Classical clustering approaches often represent students by binary enrollment vectors, which may miss relational structure from shared courses. We revisit sectioning via graph representation learning: after removing constant courses, we build a weighted student co-enrollment graph and learn low-dimensional embeddings with Node2Vec from biased random walks. Standard clustering is then applied without changing the downstream algorithm. We compare against the conventional enrollment representation, PCA with KMeans at matched dimensionality, and spectral clustering on the same graph. On six real-world courses (341 enrollments from 210 students), the primary configuration ($p = 1.0$, $q = 0.5$, $d = 2$), selected by a quality–balance Pareto analysis over 20 seeds, improves average Silhouette by about **33%** over PCA, the strongest baseline, and wins on all courses at the reference seed. Configurations that maximize Silhouette alone ($d = 1$) can yield severely imbalanced sections; the adopted setting balances separation quality and section sizes, while the quality-optimal configuration is reported as an ablation. The method is lightweight (under nine seconds per course) and can be inserted into existing sectioning pipelines without modifying clustering stages.

Keywords: Student sectioning, Graph representation learning, Node2Vec, DeepWalk, Course timetabling, Clustering

1 Introduction

Student sectioning is a classical subproblem of university course timetabling, aiming to divide enrolled students into balanced instructional sections while preserving the flexibility required for subsequent scheduling and resource allocation. The quality of sectioning can affect downstream timetabling by influencing the compatibility of student groups with available classrooms, instructors, and time slots. Consequently, student sectioning has long been studied as an important problem within educational scheduling and constraint satisfaction.

Research on student sectioning dates back more than two decades. Early studies investigated clustering-based approaches, including fuzzy clustering with feature selection and later nature-inspired optimization methods, demonstrating the potential of data-driven grouping strategies for student sectioning [1, 2]. Other studies formulated student sectioning as an optimization or constraint programming problem within larger university timetabling systems [3–7]. Together, these studies established important computational foundations for student sectioning and highlighted its role in educational resource management.

A common representation in clustering-based approaches is a student–course enrollment matrix, in which each student is represented by a binary vector indicating the courses in which the student is enrolled. This representation provides a direct description of student–course relationships and can be effective for sectioning. However, students can also be viewed as nodes in a relational network, where students are connected according to shared course enrollments. Such a representation makes it possible to model relationships between students not only through individual shared courses, but also through their connectivity patterns in the resulting co-enrollment network.

An important question is therefore whether the relational structure of student enrollments can provide useful information beyond the original enrollment representation. This question is particularly relevant because conventional dimensionality-reduction techniques can already substantially improve clustering in high-dimensional enrollment spaces. Thus, evaluating

graph-based representations requires distinguishing their contribution from the general benefit of reducing the dimensionality of the input space.

Graph representation learning provides a natural framework for addressing this question. In particular, random-walk-based methods such as Node2Vec learn continuous node representations from the contexts generated by biased random walks on a graph [8, 9]. Node2Vec extends DeepWalk by introducing return and in-out parameters that control the balance between local and global graph exploration. These representations can encode structural relationships among nodes and provide low-dimensional features that can subsequently be used by conventional machine-learning algorithms.

Graph representation learning appears to have received limited attention in the student sectioning literature, despite the natural relational structure of student co-enrollment data. This motivates the central question addressed in this paper:

Can graph representation learning provide student representations that improve sectioning quality beyond both conventional enrollment representations and standard dimensionality reduction?

To investigate this question, we model student enrollments as a weighted co-enrollment graph and learn student embeddings using Node2Vec [9], a random-walk-based graph representation learning method that extends DeepWalk [8] through biased random walks controlled by return and in-out parameters. The learned embeddings are then supplied to standard clustering algorithms for section formation. Importantly, the clustering stage is kept unchanged, allowing the effect of the learned representation to be examined independently of the choice of clustering algorithm. To assess whether the observed improvement is attributable merely to dimensionality reduction, we compare the proposed representation with a PCA-based representation having the same dimensionality. We also include spectral clustering as a graph-based baseline.

Experimental results on six real-world university courses comprising 341 student-course enrollments from 210 students show that Node2Vec provides substantially better clustering quality than the conventional enrollment representation and

spectral clustering. PCA also produces a considerable improvement, but Node2Vec achieves higher average Silhouette Scores than PCA followed by KMeans, indicating that graph-based representations capture useful structural information beyond linear dimensionality reduction. Additional experiments include a quality–balance Pareto analysis for selecting the primary Node2Vec configuration, parameter sensitivity, clustering stability across 20 random seeds, computational cost, and the resulting low-dimensional representations.

The main contributions of this paper are summarized as follows.

1. We investigate the application of graph representation learning to the student sectioning problem by constructing a weighted student co-enrollment graph and learning student representations using Node2Vec, a random-walk-based method with controllable exploration bias. While Node2Vec is an established technique in network analysis, its application to student sectioning via weighted co-enrollment graphs has received limited attention, and we demonstrate substantial improvements over conventional approaches.
2. We provide a crucial ablation-like analysis by systematically comparing graph-based representations with both the conventional student–course enrollment representation and a PCA-based dimensionality-reduction baseline. This analysis separates the effect of generic dimensionality reduction (PCA) from actual structural graph learning (Node2Vec), proving that the structural information captured by random walks provides additional value beyond just compressing data. Spectral clustering is additionally evaluated as a direct graph-based clustering baseline.
3. We empirically demonstrate on six real-world university courses that Node2Vec consistently outperforms both the conventional representation and PCA-based dimensionality reduction across all evaluated courses. Due to the limited number of course-specific sectioning instances ($n = 6$) available from a single institutional dataset, we focus on effect sizes and consistency of improvement rather than formal statistical hypothesis testing, providing strong evidence for practical significance.

4. We provide comprehensive empirical analyses of the quality–balance trade-off underlying the selection of the primary Node2Vec configuration, parameter sensitivity, clustering stability across 20 random seeds, practical significance through effect sizes and consistency, runtime, and embedding structure, providing a detailed characterization of the proposed framework.
5. We show that graph-based student representations can be seamlessly incorporated into existing clustering-based sectioning pipelines without modifying the downstream clustering algorithms, providing a simple and practical framework for improving student groupings used in subsequent timetabling processes.

The remainder of this paper is organized as follows. Section 2 reviews previous work on student sectioning and graph representation learning. Section 3 presents the proposed framework. Section 4 reports the experimental results and comparative analyses. Section 5 discusses the findings, while Section 6 concludes the paper.

2 Related Work

This work revisits the classical student sectioning problem from the perspective of graph representation learning. The related literature is therefore reviewed from two complementary directions: classical approaches to student sectioning and course timetabling, and graph representation learning for relational data. We then position the present study at the intersection of these two directions.

2.1 Student Sectioning and Course Timetabling

Student sectioning is a fundamental component of university course timetabling and educational scheduling. Its objective is to partition students into instructional sections while satisfying scheduling constraints and making efficient use of institutional resources. Because section assignments affect the flexibility of subsequent timetable construction, student sectioning has long been studied as an optimization problem within operations research and constraint programming.

Early studies primarily formulated student sectioning as part of larger timetabling systems.

Aubin and Ferland [3] incorporated section generation into a university timetabling framework, while Hertz and Robert [4] decomposed the scheduling process into assignment subproblems including student sectioning. Carter [5] considered grouping students according to course selections before timetable construction with the objective of reducing subsequent scheduling conflicts. Other approaches employed tabu search [6] and priority-based constraint satisfaction techniques [10]. These studies established student sectioning as an important component of integrated university timetabling systems.

Data-driven approaches subsequently introduced clustering techniques for student sectioning. Fuzzy clustering with feature selection was proposed in [1], representing students through enrollment-based feature vectors and applying fuzzy c-means clustering together with criteria for evaluating section balance. A Fish School clustering algorithm was later developed specifically for student sectioning, demonstrating the applicability of nature-inspired optimization to this problem [2]. Although these approaches differ in their optimization mechanisms, they generally represent students through their course enrollment information and subsequently apply a clustering or partitioning procedure.

2.2 Graph Representation Learning

Graph representation learning provides an alternative way of representing objects when their relationships are an important source of information. Instead of treating observations as independent feature vectors, graph embedding methods exploit graph topology to learn continuous low-dimensional representations that reflect structural relationships among nodes.

DeepWalk [8] introduced a widely adopted random-walk-based approach to graph representation learning by generating truncated random walks and applying the Skip-Gram model to the resulting node sequences. The learned embeddings capture patterns of node co-occurrence induced by the graph structure. Node2Vec [9] subsequently extended this framework through biased random walks that provide greater control over local and global graph exploration. Related studies have investigated weighted random walks and other graph sampling strategies [11]. Graph

representation and graph-based learning techniques have also been employed in a variety of application-oriented problems, including network embedding, structured optimization, and graph neural network-based anomaly detection [12–14].

These developments illustrate the broader applicability of graph-based learning beyond conventional graph benchmarks. In particular, they suggest that exploiting relational structure can be useful when the relationships among entities contain information that is difficult to capture directly from their original feature descriptions. Broader surveys document the rapid development of graph representation learning, including both classical embedding methods and graph neural networks [15]. In educational data mining, deep learning techniques are increasingly applied to student modeling, performance prediction, and recommendation [16], but these studies typically do not address balanced instructional sectioning from co-enrollment structure.

2.3 Graph Representation Learning for Student Sectioning

The student–course enrollment data used in sectioning naturally induces a relational structure among students. Two students can be connected according to the courses they jointly attend, resulting in a student co-enrollment graph. Such a graph provides an alternative representation of the same underlying enrollment information and makes it possible to investigate whether structural patterns in the co-enrollment network can improve student grouping.

To our knowledge, the use of graph representation learning as an explicit representation stage for student sectioning has received little attention. The present work investigates this intersection by constructing a weighted student co-enrollment graph and learning student embeddings using Node2Vec. Node2Vec is a foundational method in random-walk-based graph representation learning: it extends DeepWalk with explicit control over local and global exploration, is lightweight on small institutional graphs, and is directly compatible with off-the-shelf clustering algorithms. The resulting embeddings are subsequently supplied to standard clustering algorithms, without modifying the clustering procedure itself.

An important aspect of the present study is that the improvement is not evaluated only against the original high-dimensional enrollment representation. Since dimensionality reduction can itself substantially affect clustering performance, we additionally consider PCA followed by KMeans as a representation-based baseline with the same embedding dimensionality as Node2Vec. We also evaluate spectral clustering directly on the co-enrollment graph as a graph-based clustering baseline. These comparisons allow us to examine whether the observed benefit of Node2Vec is attributable merely to dimensionality reduction or whether graph-based representation learning provides additional useful information for student sectioning.

The present study therefore complements earlier clustering-based approaches to student sectioning rather than replacing them. Its central objective is to examine whether a Node2Vec-based representation learning stage can improve the quality of student representations while leaving the subsequent clustering pipeline unchanged.

3 Methodology

The proposed framework revisits the student sectioning problem from a graph representation learning perspective. Instead of representing students solely through a binary student-course enrollment matrix, we construct a student co-enrollment graph and learn continuous vector representations of students from the topology of this graph. These representations are subsequently used as input to conventional clustering algorithms to generate student sections.

The overall framework consists of four main stages: (1) construction of the student co-enrollment graph, (2) graph representation learning using random walks, (3) clustering of the learned student representations, and (4) evaluation of the resulting student sections. Figure 1 illustrates the overall workflow.

3.1 Student Co-enrollment Graph

Let $S = \{s_1, s_2, \dots, s_n\}$ denote the set of students and let $C = \{c_1, c_2, \dots, c_m\}$ denote the set of courses. Student enrollment information can be represented by a binary matrix $X \in \{0, 1\}^{n \times m}$, where

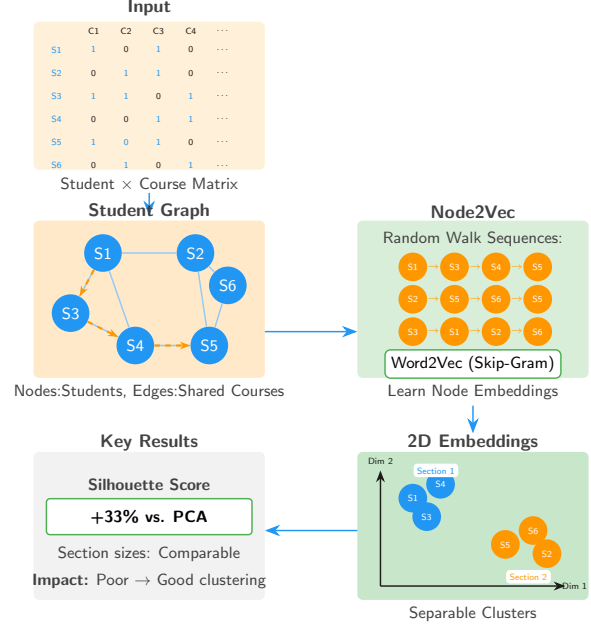


Fig. 1: Overview of the proposed framework. A student-course enrollment matrix is transformed into a student co-enrollment graph, from which random walks are generated and fed to a Skip-Gram (Word2Vec) model to learn low-dimensional student embeddings. The learned embeddings are then clustered to produce student sections.

$$X_{ic} = \begin{cases} 1, & \text{if student } s_i \text{ is enrolled in course } c, \\ 0, & \text{otherwise.} \end{cases}$$

Although the enrollment matrix provides a direct representation of student-course relationships, it does not explicitly represent relationships between students. We therefore transform the enrollment information into a student co-enrollment graph.

We define an undirected weighted graph $G = (V, E, W)$, where each vertex $v_i \in V$ corresponds to a student s_i . An edge $(v_i, v_j) \in E$ is established when students s_i and s_j share at least one enrolled course. The edge weight measures the strength of their co-enrollment relationship.

In the basic version of the proposed framework, the edge weight between students s_i and s_j is defined as the number of courses in which they are simultaneously enrolled:

$$w_{ij} = \sum_{c \in C} X_{ic} X_{jc}.$$

Consequently, students who share several courses are connected by stronger edges, whereas students who share only one course have a weaker connection. This weighted representation preserves more information than a binary adjacency matrix and provides a natural basis for subsequent random-walk-based representation learning.

The resulting graph can be represented by a weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$, where $W_{ij} = w_{ij}$ for connected students and $W_{ij} = 0$ otherwise. Because the co-enrollment relationship is symmetric, the resulting graph is undirected and $W_{ij} = W_{ji}$.

3.2 Preprocessing: Removal of Constant Features

Before constructing any representation, we apply a standard preprocessing step to the student-course enrollment matrix. Courses that are taken by every student in a sectioning instance have zero variance and provide no discriminative information for clustering. Such constant features are therefore removed from the enrollment matrix before any representation or graph is constructed.

Specifically, for each target course, we identify courses c for which $X_{ic} = 1$ for all students s_i in the student set. These universally shared courses are removed, and the remaining enrollment matrix is used for all subsequent steps: BoW representation, PCA projection, graph construction, and random-walk-based embedding. This ensures that all methods operate on the same preprocessed data.

In the six sectioning instances studied here, exactly one constant course is identified and removed per instance. The resulting graphs are non-complete, with average density approximately 0.72, in contrast to the fully connected graphs that arise when the constant course is included.

3.3 Random-walk-based Student Representation Learning

After constructing the co-enrollment graph, we learn a vector representation for each student using Node2Vec [9], a random-walk-based graph embedding method. The main idea is to treat the

graph analogously to a corpus: students correspond to words, and random walks correspond to sentences. Students that frequently occur in similar graph contexts are therefore expected to obtain similar vector representations.

Node2Vec generates biased random walks in which the transition probability depends on both the previous and the next-to-previous node in the walk. Let u denote the previous node and v denote the current node. The unnormalized transition probability to a neighbor x is defined as

$$\pi(v, x) = \alpha_{pq}(u, x) \cdot w_{vx},$$

where w_{vx} is the edge weight and the bias factor $\alpha_{pq}(u, x)$ is

$$\alpha_{pq}(u, x) = \begin{cases} 1/p, & \text{if } d(u, x) = 0, \\ 1, & \text{if } d(u, x) = 1, \\ 1/q, & \text{if } d(u, x) = 2, \end{cases}$$

where $d(u, x)$ denotes the shortest-path distance between nodes u and x . The parameter p controls the likelihood of returning to the previous node (return parameter), and q controls whether the walk explores locally (BFS-like, $q < 1$) or globally (DFS-like, $q > 1$). Setting $p = 1$ and $q = 1$ recovers the standard unbiased random walk used by DeepWalk.

Thus, students with stronger co-enrollment relationships have a higher probability of being visited during the random walk, and the bias parameters allow the walk to emphasize either local neighborhood structure or broader graph exploration. Repeating this procedure produces a collection of student sequences (Algorithm 2),

$$\mathcal{W} = \{W_1, W_2, \dots, W_R\},$$

where R denotes the total number of generated walks.

The resulting walks capture both local and higher-order relationships in the student graph (Figure 2 illustrates an example walk). For example, two students who do not directly share a course may nevertheless occur frequently in similar random-walk contexts because they are connected through common groups of students. Such relationships cannot be directly captured by treating each student’s enrollment vector independently.

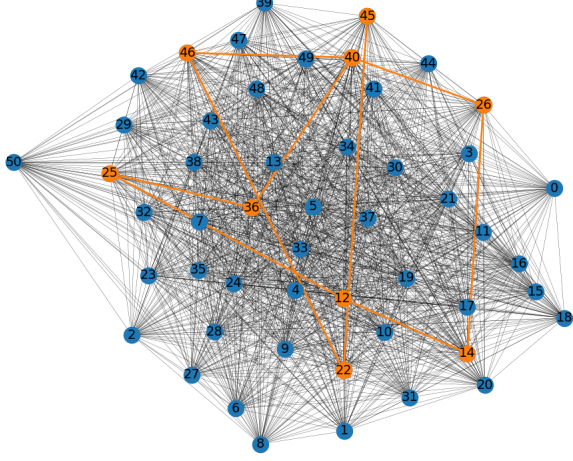


Fig. 2: Example random walk on the student graph for Course 2, starting at node 40 and ending at node 45: $\{40, 36, 25, 12, 14, 26, 40, 46, 22, 45\}$. Students who share courses are connected, and the walk explores the local neighborhood structure.

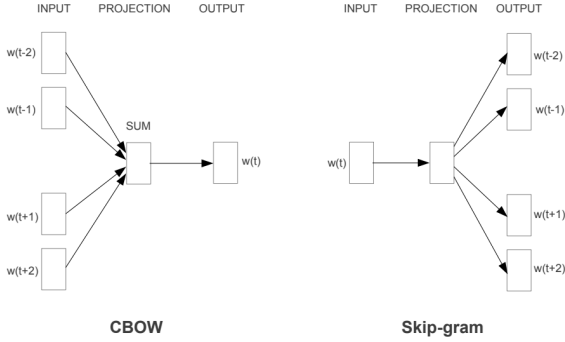


Fig. 3: The CBOW architecture predicts the current word based on the context, and the Skip-gram predicts surrounding words given the current word [17]. In Node2Vec, nodes play the role of words and random-walk sequences play the role of sentences.

3.4 Learning Student Embeddings

The generated random walks are used to learn low-dimensional student representations using a skip-gram objective. Let $z_v \in \mathbb{R}^d$ denote the embedding of student v , where d is the embedding dimension. For a student v , the objective encourages students appearing within a context window of v to have similar representations. Figure 3 illustrates the CBOW and Skip-gram architectures used in this learning stage.

Algorithm 1 Student Representation Learning using Node2Vec

Require: graph $G(V, E, W)$ constructed from student-course matrix, where students are its nodes. The input parameters are:

- return parameter p
- in-out parameter q
- walk length L
- number of walks per node γ
- embedding size d
- window size w
- epochs ϵ

Ensure: matrix of vertex (student) representations $\Phi \in \mathbb{R}^{|V| \times d}$

- 1: $\mathcal{W} \leftarrow \text{GenerateRandomWalks}(G, L, \gamma, p, q) \triangleright$ Algorithm 2
 - 2: $\Phi \leftarrow \text{TrainWord2Vec}(\mathcal{W}, d, w, \epsilon) \triangleright$ Sect. 3.4
 - 3: **return** Φ
-

Given a context window of size r , the optimization objective can be expressed as

$$\max_{\{z_v\}} \sum_{v \in V} \sum_{u \in C(v)} \log P(u | v),$$

where $C(v)$ denotes the set of students occurring within the specified context window of v in the generated random walks.

The conditional probability can be modeled using the softmax function:

$$P(u | v) = \frac{\exp(z_u^\top z_v)}{\sum_{x \in V} \exp(z_x^\top z_v)}.$$

In practice, negative sampling can be used to make the optimization computationally efficient. The resulting embedding matrix

$$Z = \begin{bmatrix} z_1^\top \\ z_2^\top \\ \vdots \\ z_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}$$

provides a continuous representation of all students.

The important distinction from the original enrollment representation is that each vector z_i is learned from the structural context of student s_i in the entire co-enrollment network. Consequently, the representation can capture indirect and higher-order similarities between students.

Algorithm 1 summarizes the overall procedure: biased random walks are generated from the co-enrollment graph (line 1), and the Skip-Gram model is trained on these walks to produce the student embedding matrix Φ (line 2 of algorithm 1).

3.5 Student Sectioning by Clustering

Once the student embeddings have been learned, student sectioning is formulated as a clustering problem. Let k denote the desired number of sections. A clustering algorithm partitions the set of student embeddings

$$Z = \{z_1, z_2, \dots, z_n\}$$

into k clusters,

$$S = \{S_1, S_2, \dots, S_k\},$$

where each cluster S_j corresponds to a student section.

The proposed framework is deliberately independent of a particular clustering algorithm. This allows us to investigate whether the improvement comes from the learned graph representations rather than from a specific clustering technique. In our experiments, we therefore evaluate several widely used clustering methods, including K-means, Gaussian mixture models, and hierarchical (agglomerative) clustering.

For K-means, for example, the clustering objective is

$$\min_{\{S_j\}_{j=1}^k} \sum_{j=1}^k \sum_{z_i \in S_j} \|z_i - \mu_j\|_2^2,$$

where μ_j denotes the centroid of section S_j .

The resulting clusters define the student sections used in the subsequent evaluation. A simple section balance metric $B = \min(|S_1|, |S_2|) / \max(|S_1|, |S_2|)$ for two sections (or the analogous ratio for $k > 2$) can further characterize the practical suitability of the resulting sections, although a comprehensive evaluation of sectioning quality in the context of downstream timetabling constraints is beyond the scope of the present study.

3.6 Baseline Representation

To determine whether graph representation learning provides an advantage over conventional student representations, we establish a baseline based directly on the student–course enrollment matrix.

For each student s_i , the corresponding row of the enrollment matrix,

$$x_i = (X_{i1}, X_{i2}, \dots, X_{im}),$$

is used as the student representation. Thus, students are represented directly by their binary course enrollment patterns without constructing a graph or learning an embedding.

The same clustering algorithms are then applied to these enrollment vectors. This provides a controlled comparison because the clustering stage remains unchanged and the principal difference between the two approaches is the representation used to describe students.

Accordingly, the comparison considers two alternative pipelines:

Enrollment representation $\rightarrow$ Clustering, and

Graph representation $\rightarrow$ Embedding $\rightarrow$ Clustering.

This design allows the contribution of graph representation learning to be evaluated independently of the choice of clustering algorithm.

3.7 Hyperparameter Configuration

The graph embedding procedure contains several hyperparameters that may affect the quality of the learned representations. The principal parameters considered in this study include the embedding dimension d , the random-walk length L , the number of walks generated per student γ , and the context-window size w .

The embedding dimension controls the dimensionality of the learned student representations. A larger value provides greater representational capacity but may also increase computational cost and the risk of overfitting. The walk length determines how far the random walks explore the co-enrollment network, while the number of walks controls the amount of structural context available during training. The context-window size determines the range of neighboring positions considered when learning the embedding.

Rather than relying on a single arbitrary configuration, we conduct a comprehensive grid

search over the Node2Vec-specific parameters p , q , and d (Section 3.8), as well as sensitivity experiments over the remaining hyperparameters. The purpose is not to optimize the method specifically for a particular dataset, but to examine whether the observed improvement is robust across reasonable parameter settings.

3.8 Node2Vec Grid Search

To identify the optimal Node2Vec configuration for student sectioning, we conduct a comprehensive grid search over the key hyperparameters that control the random-walk behavior and embedding dimensionality. Specifically, we evaluate all combinations of:

- Return parameter $p \in \{0.5, 1.0, 2.0\}$
- In-out parameter $q \in \{0.5, 1.0, 2.0\}$
- Embedding dimension $d \in \{1, 2, 3, 5, 10\}$

This yields $3 \times 3 \times 5 = 45$ unique configurations. Each configuration is evaluated on all six courses with a fixed random seed (seed = 42) to ensure reproducibility. For each course and configuration, we generate Node2Vec embeddings using $L = 10$ walk length, $\gamma = 80$ walks per node, window size $w = 5$, and $\epsilon = 30$ training epochs, followed by KMeans clustering with $k = 2$ clusters. The primary evaluation metric is the average Silhouette Score across all six courses.

The grid search results are summarized in Figure 4, which shows heatmaps of the average Silhouette Score for each (p, q) combination at each embedding dimension d . Each heatmap uses an independent color scale to facilitate visual comparison of the variation within each dimension.

The grid search reveals several important findings:

1. **Embedding dimension critically affects performance.** The best results are consistently achieved at $d = 1$, with average Silhouette Scores ranging from 0.732 to 0.765. Performance degrades substantially as d increases beyond 2, with $d = 10$ yielding average Silhouette Scores below 0.4 for most configurations. This suggests that the co-enrollment graph structure provides sufficient discriminatory information for student sectioning in a very low-dimensional space.

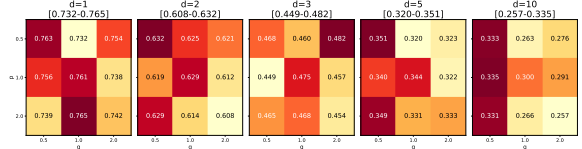


Fig. 4: Heatmaps of average Silhouette Score for the Node2Vec (p, q, d) grid search. Each panel shows the mean Silhouette Score across six courses for a fixed embedding dimension d , with rows corresponding to the return parameter p and columns to the in-out parameter q . The silhouette-optimal configuration ($p = 2.0, q = 1.0, d = 1$, avg. Silhouette = 0.765) is highlighted in the top-left panel; Section 4.2 shows that this quality-optimal configuration yields severely imbalanced sections, and selects the primary configuration at $d = 2$ through a quality–balance analysis.

Table 1: Top 5 Node2Vec configurations from the grid search, ranked by average Silhouette Score across six courses.

Rank	p	q	d	Avg. Silhouette
1	2.0	1.0	1	0.7654
2	0.5	0.5	1	0.7634
3	1.0	1.0	1	0.7613
4	1.0	0.5	1	0.7557
5	0.5	2.0	1	0.7535

2. **The return parameter p interacts with embedding dimension.** At $d = 1$, higher values of p (reducing the likelihood of returning to the previous node) tend to produce better clustering. The best configuration overall is $p = 2.0, q = 1.0, d = 1$ with an average Silhouette Score of 0.765. At higher dimensions, the effect of p is less pronounced.
3. **The in-out parameter q has a smaller effect.** Across all dimensions, varying q between 0.5 and 2.0 produces relatively small changes in Silhouette Score compared to varying p or d . The neutral value $q = 1.0$ (corresponding to unbiased random walks) performs competitively across all dimensions.

Table 1 presents the top 5 configurations. The best configuration ($p = 2.0, q = 1.0, d = 1$) achieves an average Silhouette Score of 0.7654, with per-course scores of 0.963, 0.593, 0.927, 0.634,

0.838, and 0.613 for courses 1–6 respectively. The second-best configuration ($p = 0.5$, $q = 0.5$, $d = 1$) achieves 0.7634, demonstrating that multiple configurations at $d = 1$ perform competitively.

If clustering quality were the only criterion, the best configuration $p = 2.0$, $q = 1.0$, $d = 1$ would be adopted. Student sectioning, however, also requires reasonably balanced section sizes. As quantified in Section 4.2, the silhouette-optimal $d = 1$ configurations produce severely imbalanced sections (e.g., splits of 72–2 and 47–1 students in courses 1 and 3), whereas $d = 2$ configurations attain a markedly better quality–balance trade-off. We therefore complement the grid search with an explicit multi-objective (Pareto) analysis over clustering quality and section balance, validated over 20 random seeds; it selects $p = 1.0$, $q = 0.5$, $d = 2$ as the primary configuration used in the core experiments, while the silhouette-optimal and neutral configurations are analyzed separately as ablations (Section 4.7).

All grid search results, including per-course Silhouette Scores, cluster labels, and embeddings for key configurations, are publicly available in the accompanying repository to ensure full reproducibility.

3.9 Evaluation of Student Sectioning

The quality of the resulting student sections is evaluated using clustering-based internal validity measures. Since the primary objective is to create coherent groups of students, the Silhouette Score is used as the main evaluation metric.

For a student i , let $a(i)$ denote the average distance between i and the other students in the same section, and let $b(i)$ denote the minimum average distance between i and the students belonging to another section. The silhouette coefficient of student i is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

The overall Silhouette Score is obtained by averaging this coefficient over all students:

$$S = \frac{1}{n} \sum_{i=1}^n s(i).$$

The Silhouette Score ranges from -1 to 1 , with larger values indicating better within-section cohesion and between-section separation.

To complement the primary evaluation, we also examine clustering stability across multiple random seeds. In particular, the Adjusted Rand Index (ARI) is used to quantify the agreement between clustering solutions obtained under different random initializations. High ARI values indicate that the learned representation leads to stable section assignments rather than highly variable partitions.

Finally, we report effect-size measures to quantify the practical magnitude of the observed improvements, focusing on consistency across courses and the magnitude of improvement rather than formal statistical hypothesis testing.

4 Experimental Results

We evaluate the proposed graph representation learning framework for student sectioning using a real-world university dataset and a series of experiments designed to examine both its effectiveness and robustness. The experiments address four main questions: (i) whether graph-based representations improve student clustering compared with the conventional student-course representation, (ii) whether the improvement can be explained solely by dimensionality reduction, (iii) whether direct graph clustering provides comparable benefits, and (iv) how sensitive the proposed approach is to its main hyperparameters and stochastic components.

4.1 Dataset and Experimental Setup

We use the real-world student-course enrollment dataset introduced in [1], which contains 210 students and 38 courses. To evaluate the student sectioning problem, we select six relatively large courses comprising 341 student-course enrollments (74, 51, 48, 49, 52, and 67 enrollments per course, respectively). These courses provide a range of realistic sectioning scenarios while retaining sufficient enrollment information to construct meaningful student co-enrollment graphs.

The proposed method constructs a weighted student co-enrollment graph from the preprocessed enrollment data and learns student representations using Node2Vec random walks and the Skip-Gram model. The resulting embeddings are subsequently provided to a clustering algorithm for student sectioning. Unless otherwise stated,

KMeans is used as the downstream clustering algorithm.

As the conventional baseline, we use the original binary student-course enrollment representation (after constant-feature removal), referred to as the *Student-Course Representation* or *BoW* representation. To examine whether the observed improvement is primarily attributable to dimensionality reduction, we additionally evaluate a PCA-based baseline in which the conventional representation is projected to the same two-dimensional space used by the graph embedding methods before applying KMeans. We also evaluate spectral clustering directly on the student co-enrollment graph as a graph-based baseline that does not use an intermediate learned embedding. Spectral clustering uses the precomputed weighted co-enrollment affinity with $k=2$ clusters and k -means assignment of the spectral embedding.

We determine the Node2Vec configuration through a comprehensive grid search over the return parameter $p \in \{0.5, 1.0, 2.0\}$, the in-out parameter $q \in \{0.5, 1.0, 2.0\}$, and the embedding dimension $d \in \{1, 2, 3, 5, 10\}$, conducted on all six courses with seed 42 (see Section 3.8 for details). The grid search shows that silhouette-optimal configurations concentrate at $d = 1$, with $p = 2.0$, $q = 1.0$, $d = 1$ achieving the highest average Silhouette Score of 0.7654. Because maximizing clustering quality alone can produce impractically imbalanced sections, we additionally conduct a quality–balance Pareto analysis with 20-seed validation (Section 4.2), which selects $p = 1.0$, $q = 0.5$, $d = 2$ as the primary configuration used in the core experiments. This configuration is matched in dimensionality with the PCA baseline and supports direct 2D visualization of the embeddings. The silhouette-optimal configuration ($p = 2.0$, $q = 1.0$, $d = 1$) and the neutral DeepWalk-equivalent configuration ($p = 1.0$, $q = 1.0$, $d = 2$) are analyzed separately as ablations (Section 4.7). Because configuration selection and performance evaluation use the same six sectioning instances, the reported scores should be interpreted as in-sample estimates on this dataset; multi-institutional held-out evaluation remains future work.

Source code and the dataset used in the experiments are publicly available at <https://github.com/mamintoosi/Node2Vec-SSP>.

The primary evaluation metric is the Silhouette Score [18], which considers both intra-cluster cohesion and inter-cluster separation. Higher values indicate better-defined clusters. We additionally report the Davies–Bouldin Index (DBI), for which lower values are better, and the Calinski–Harabasz (CH) Index, for which higher values are better.

4.2 Quality–Balance Trade-off and Primary Configuration Selection

The grid search ranks configurations by a single objective (average Silhouette Score), whereas practical student sectioning must also produce sections of usable size. We therefore treat configuration selection as a bi-objective problem, combining the average Silhouette Score with the average section balance $B = \min(|S_1|, |S_2|) / \max(|S_1|, |S_2|)$ computed from the resulting cluster assignments. A configuration is Pareto-optimal if no other configuration is at least as good in both objectives and strictly better in one.

The analysis of all 45 grid configurations reveals a sharp conflict between the two objectives. Every $d = 1$ configuration collapses courses 1 and 3 into near-singleton sections (sizes 72/2 and 47/1, $B \approx 0.02$ –0.03), and no configuration with an average Silhouette Score of at least 0.7 reaches an average balance of 0.6. The balanced region of the frontier lies at $d = 2$. We therefore shortlist three representative candidates and validate them over 20 random seeds (random walks, Word2Vec training, and KMeans are regenerated for each seed): the quality-optimal configuration ($p = 2.0$, $q = 1.0$, $d = 1$), the neutral DeepWalk-equivalent configuration ($p = 1.0$, $q = 1.0$, $d = 2$), and the balance-optimal configuration at $d \leq 2$ ($p = 1.0$, $q = 0.5$, $d = 2$). Table 2 reports the results.

Three observations follow from Table 2. First, the quality-optimal configuration achieves the highest Silhouette Score but the worst and most variable section balance, including the degenerate splits noted above, and the lowest stability. Second, the balance-optimal configuration ($p = 1.0$,

Table 2: Multi-seed validation of the three shortlisted Node2Vec configurations. Values are averaged over the six courses. “20 seeds” refers to full-pipeline repetitions (new random walks, Word2Vec training, and KMeans per seed); $\pm$ denotes the average within-course standard deviation across the 20 seeds. Sil.: average Silhouette Score ($\uparrow$); B : average section balance ($\uparrow$); min-course B : the lowest per-course average balance ($\uparrow$); ARI: Adjusted Rand Index over 20 KMeans initializations on the fixed seed-42 embeddings ($\uparrow$). The adopted primary configuration is listed last.

Config (p, q, d)	Sil. (seed 42)	Sil. (20 seeds)	B (seed 42)	B (20 seeds)	min-course B	ARI
(2.0, 1.0, 1), quality-optimal	0.765	0.755 $\pm$ 0.037	0.308	0.406	0.021	0.980
(1.0, 1.0, 2), neutral	0.629	0.618 $\pm$ 0.012	0.558	0.606	0.292	0.984
(1.0, 0.5, 2), balance-optimal	0.619	0.629 $\pm$ 0.018	0.652	0.602	0.349	0.994

$q = 0.5$, $d = 2$) dominates the neutral configuration in average Silhouette over 20 seeds (0.629 vs. 0.618), worst-course balance (0.349 vs. 0.292), and stability (KMeans ARI 0.994 vs. 0.984), while the two are virtually tied in average balance (0.602 vs. 0.606). Third, single-seed (seed 42) rankings differ from the 20-seed means (e.g., the neutral configuration scores 0.629 at seed 42 but 0.618 on average), confirming that configuration selection should rely on multi-seed evidence. Based on this analysis, we adopt ($p = 1.0$, $q = 0.5$, $d = 2$) as the primary configuration for the core experiments; the quality-optimal and neutral configurations are examined further in the ablation study (Section 4.7).

4.3 Overall Clustering Performance and Baseline Comparison

We compare the Node2Vec embeddings with the conventional student-course representation, PCA+KMeans, and spectral clustering to determine whether the observed improvement can be attributed simply to dimensionality reduction, to the specific random-walk strategy, or to direct exploitation of graph structure. Our primary Node2Vec configuration is the Pareto-selected $p = 1.0$, $q = 0.5$, $d = 2$ (Section 4.2), which provides a matched-dimensionality comparison with the PCA baseline and supports direct 2D visualization. The silhouette-optimal and neutral configurations are analyzed separately in Section 4.7.

Figure 5 and Table 3 summarize the Silhouette scores obtained by the four approaches across the six selected courses.

The primary Node2Vec configuration ($p = 1.0$, $q = 0.5$, $d = 2$) achieves an average Silhouette Score of 0.619, which substantially

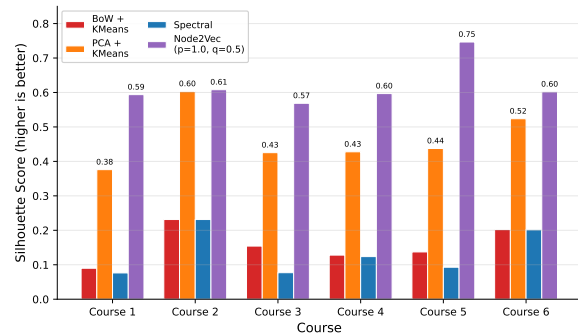


Fig. 5: Silhouette scores ($\uparrow$ higher is better) for four student sectioning methods across six courses: BoW + KMeans, PCA + KMeans, Spectral Clustering, and Node2Vec under the primary Pareto-selected configuration ($p = 1.0$, $q = 0.5$, $d = 2$).

exceeds all baseline methods. This represents a relative improvement of approximately 33% over PCA+KMeans (average 0.466), the strongest baseline. Because the PCA baseline operates in the same two-dimensional space, this comparison isolates the contribution of the learned graph representation from that of dimensionality reduction itself.

The consistent superiority of Node2Vec over PCA at matched dimensionality confirms that the graph-based representation learning captures structural information beyond what is achievable through dimensionality reduction alone. The silhouette-optimal configuration at $d = 1$ reaches a higher average Silhouette Score of 0.765 (Section 3.8), but, as the quality–balance analysis shows, this comes at the cost of severely imbalanced sections; it is therefore examined as

Table 3: Silhouette Score ($\uparrow$ higher is better) comparison across four methods for student sectioning. The primary Node2Vec configuration is the Pareto-selected $p = 1.0$, $q = 0.5$, $d = 2$ (Section 4.2), which operates in the same dimensionality as the PCA baseline.

Course	BoW + KMeans	PCA + KMeans	Spectral	Node2Vec ($p=1.0, q=0.5, d=2$)
1	0.090	0.376	0.083	0.594
2	0.231	0.603	0.205	0.608
3	0.154	0.425	0.135	0.569
4	0.128	0.428	0.128	0.597
5	0.137	0.437	0.074	0.746
6	0.202	0.524	0.143	0.602
Average	0.157	0.466	0.128	0.619

an ablation rather than adopted as the primary setting.

Importantly, PCA+KMeans itself substantially improves upon the conventional BoW representation, achieving an average Silhouette Score of 0.466. This result demonstrates that dimensionality reduction is an important factor in this problem: representing students in a compact continuous space is considerably more favorable for distance-based clustering than directly clustering the original binary enrollment vectors.

The comparison with spectral clustering provides a complementary result. Spectral clustering achieves an average Silhouette Score of 0.128, which is lower than the conventional BoW+KMeans baseline (0.157) and substantially worse for most courses. Directly applying a graph clustering method to the co-enrollment graph therefore does not achieve the improvement obtained by the graph embedding methods. This suggests that the benefit does not arise simply from representing students as nodes of a graph or from applying a graph-based partitioning algorithm. Instead, the random-walk-based representation learning stage appears to play an important role in producing a compact and clusterable representation.

Node2Vec with $p = 1$ and $q = 1$ is theoretically equivalent to the unbiased random walk used by DeepWalk [8]; this neutral configuration achieves an average Silhouette Score of 0.629 at $d = 2$ (seed 42) and is analyzed as an ablation in Section 4.7. The configuration analysis further shows that both the embedding dimension and the walk bias parameters measurably affect clustering

Table 4: Student co-enrollment graph statistics after constant-feature removal. n is the number of students, m is the number of remaining courses, $|E|$ is the number of edges, and $\bar{d}$ is the average weighted degree.

Course	n	m	$ E $	Density	Components
1	74	34	1,828	0.677	3
2	51	33	866	0.679	1
3	48	27	894	0.793	2
4	49	27	1,043	0.887	1
5	52	31	912	0.688	1
6	67	32	1,325	0.599	1
Avg.	56.8	30.7	1,145	0.721	—

quality, with a mild local bias ($q = 0.5$) providing the best quality–balance trade-off over repeated seeds.

Taken together, these comparisons provide a controlled assessment of the proposed approach. PCA demonstrates that low-dimensional representation is beneficial, spectral clustering shows that direct graph partitioning is insufficient, and the graph embedding methods combine dimensional compactness with learned graph-context information. The results therefore support the interpretation that the performance gain is not solely a consequence of dimensionality reduction, but also reflects information captured from the relational structure of the student co-enrollment graph.

Table 4 summarizes the graph structure after constant-feature removal.

Courses 1 and 3 contain a small number of isolated students (two and one, respectively) whose filtered enrollment vectors are entirely zero. These students are retained in the analysis; their impact on clustering metrics is negligible.

4.4 Section Balance Analysis

To characterize the practical suitability of the resulting sections, we compute the section balance metric $B = \min(|S_1|, |S_2|) / \max(|S_1|, |S_2|)$ for the two sections produced by each method with KMeans clustering (or spectral clustering for the Spectral baseline). Table 5 compares the section balance across all four methods.

The primary Pareto-selected configuration produces the most balanced sections overall, with an average balance of 0.652 compared with 0.465–0.471 for the baselines, and no course below $B = 0.368$. This confirms that the configuration selection of Section 4.2 does not trade balance for quality: relative to all baselines, the primary configuration simultaneously improves both objectives. Course 5 remains the most challenging sectioning scenario across all methods ($B = 0.368$ –0.405). In practice, an explicit balancing constraint could still be incorporated into the clustering stage when institutional policies require tighter size control; the quality-optimal configuration analyzed in Section 4.7, in contrast, produces degenerate sections without such a constraint.

4.5 Node2Vec Parameter Sensitivity

We conduct a parameter sensitivity experiment over $p \in \{0.5, 1.0, 2.0\}$ and $q \in \{0.5, 1.0, 2.0\}$, keeping all other parameters at their fixed reference values ($L=10$, $\gamma=80$, $d=1$, $w=5$, $\epsilon=30$); this table thus explores the quality objective at the silhouette-optimal dimension, while the quality–balance trade-off is analyzed in Section 4.2. Table 6 reports the average Silhouette Score across the six courses for each configuration.

The best configuration at $d=1$ is $p=2.0$, $q=1.0$ with an average Silhouette Score of 0.7654, which is the globally optimal configuration identified by the grid search (Section 3.8). The neutral configuration $p=1$, $q=1$ (equivalent to unbiased random walks, corresponding to DeepWalk [8]) achieves 0.7613. The sensitivity range at $d=1$ is wider (0.7317–0.7654) than at higher dimensions, and the return parameter p shows a clearer interaction

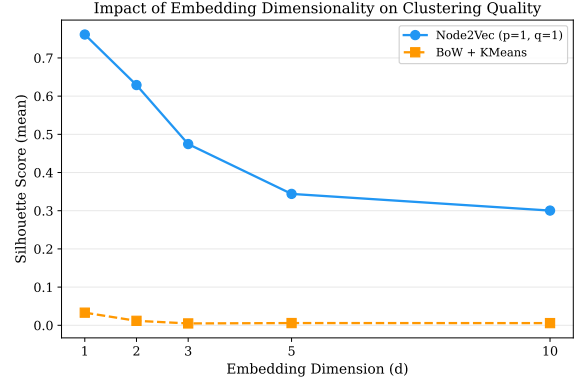


Fig. 6: Silhouette scores ($\uparrow$ higher is better) for the graph embedding method ($p=1$, $q=1$) across different embedding dimensions. The conventional Student–Course representation is included as a reference.

with performance at this low dimension. These results suggest that at $d=1$, the choice of p and q has a measurable effect on clustering quality, with higher values of p (reducing the likelihood of returning to the previous node) generally favoring better performance.

4.6 Embedding Dimension and Hyperparameter Sensitivity

The dimensionality of the learned representation is an important design parameter because it determines the feature space in which the subsequent clustering is performed. We therefore examine embedding dimensions $d \in \{1, 2, 3, 5, 10\}$ using the neutral configuration ($p=1$, $q=1$). Figure 6 shows the resulting Silhouette scores, while Table 7 reports three complementary clustering metrics.

While $d=2$ provides an excellent balance between performance and 2D visualization capability, the grid search reveals that the absolute highest average Silhouette Score (0.761) is achieved at $d=1$ with $p=1$, $q=1$, demonstrating that the structural information can be highly compressed into a single dimension. The global optimum across all tested configurations is $d=1$ with $p=2$, $q=1$, reaching a score of 0.765. At $d=2$, the Silhouette Score is 0.629, and it decreases substantially for higher dimensions. At $d=10$, the Silhouette Score falls to 0.300. It should be noted, however, that the $d=1$ scores are obtained with severely imbalanced sections (Section 4.2); the

Table 5: Section balance analysis ($B = \min(|S_1|, |S_2|) / \max(|S_1|, |S_2|)$) across four methods for student sectioning with $k=2$ sections. The primary Node2Vec configuration is the Pareto-selected $p = 1.0, q = 0.5, d = 2$ (Section 4.2). $B=1$ indicates perfectly balanced sections. Bold values indicate the highest balance per course.

Course	BoW + KMeans	PCA + KMeans	Spectral	Node2Vec ($p=1.0, q=0.5$)
1	0.423	0.480	0.542	0.609
2	0.275	0.275	0.378	0.594
3	0.548	0.548	0.655	0.714
4	0.581	0.581	0.581	0.815
5	0.405	0.368	0.182	0.368
6	0.558	0.558	0.489	0.811
Avg.	0.465	0.468	0.471	0.652

Table 6: Node2Vec parameter sensitivity: average Silhouette Score ($\uparrow$) across six courses for different (p, q) configurations at $d=1$.

q	$p = 0.5$			$p = 1.0$			$p = 2.0$		
	0.5	1.0	2.0	0.5	1.0	2.0	0.5	1.0	2.0
Silhouette	0.7634	0.7317	0.7535	0.7557	0.7613	0.7382	0.7389	0.7654	0.7424

Table 7: Effect of embedding dimension d on clustering quality for the neutral configuration ($p=1, q=1$), averaged across the six courses. Silhouette score ($\uparrow$ higher is better), Davies–Bouldin Index ($\downarrow$ lower is better), and Calinski–Harabasz Index ($\uparrow$ higher is better).

Dimension	Silhouette ($\uparrow$)	DBI ($\downarrow$)	CH ($\uparrow$)
1	0.761	0.354	295.7
2	0.629	0.532	134.9
3	0.475	0.826	60.3
5	0.344	1.230	29.7
10	0.300	1.386	18.0

$d=2$ operating point adopted as the primary configuration retains most of the separation quality while producing practically balanced sections.

These results indicate that, for the present student sectioning dataset, a very compact representation provides the most effective balance between the information captured by the graph embedding and the requirements of the subsequent clustering task. The two-dimensional representation retains a practical advantage because the learned embeddings can be directly visualized without requiring an additional dimensionality-reduction method;

Table 8: Sensitivity to walk length L (higher Silhouette is better; averaged across six courses, $d = 2, p = 1.0, q = 1.0, \gamma = 80, w = 5$).

WL (L)	5	10	20	40	80
Silhouette	0.606	0.629	0.626	0.623	0.637

accordingly, the primary configuration adopted for the core experiments is the Pareto-selected two-dimensional setting ($p = 1.0, q = 0.5, d = 2$), while the $d = 2$ neutral configuration is reported as an ablation. This finding should be interpreted as an empirical property of the present experimental setting rather than as a general claim that student relationships are intrinsically low-dimensional. The appropriate embedding dimensionality may depend on the size and structural characteristics of the underlying enrollment graph.

We next examine the sensitivity of the graph embedding method to its principal random-walk and Skip-Gram parameters. The effects of walk length L , number of walks per node γ , and context-window size w are evaluated independently, while the remaining parameters are fixed at their default values.

Table 9: Sensitivity to the number of walks per node γ ($\uparrow$ higher Silhouette is better; averaged across six courses, $d = 2$, $p = 1.0$, $q = 1.0$, $L = 10$, $w = 5$).

Walks (γ)	10	20	40	80	160
Silhouette ($\uparrow$)	0.622	0.626	0.613	0.629	0.617

Table 10: Sensitivity to context-window size w ($\uparrow$ higher Silhouette is better; averaged across six courses, $d = 2$, $p = 1.0$, $q = 1.0$, $L = 10$, $\gamma = 80$).

Window (w)	1	2	5	10	20
Silhouette ($\uparrow$)	0.669	0.619	0.629	0.637	0.632

Walk length has a noticeable effect on clustering quality, as shown in Table 8. The highest Silhouette Score is obtained with $L = 80$ (0.637), followed by $L = 10$ (0.629) and $L = 20$ (0.626), with shorter walks ($L = 5$) achieving 0.606. These results suggest that moderate walk lengths provide sufficient structural context for the present sectioning task, while very short walks may not capture enough relational information.

The number of walks per node has a smaller effect than embedding dimension, as shown in Table 9. Performance peaks at $\gamma = 80$, where the highest score (0.629) is obtained, with $\gamma = 20$ (0.626) and $\gamma = 160$ (0.617) performing slightly below. The results therefore suggest that $\gamma = 80$ provides a reasonable trade-off between sampling density and computational cost for the present dataset.

The context-window size has a smaller effect among the examined parameters, as shown in Table 10. The best result is obtained at $w = 1$ (0.669), followed by $w = 10$ (0.637). The Silhouette Score remains within the range of 0.619–0.669 across the tested values. Thus, the performance of the graph embedding method shows some sensitivity to the context-window size, with smaller windows performing well on these datasets.

Overall, the sensitivity experiments show that embedding dimensionality has the largest effect on clustering quality, while walk length, number

of walks, and context-window size have comparatively smaller effects. The Silhouette Score varies from 0.300 to 0.7654 across the tested embedding dimensions, compared with a range of about 0.031 across walk lengths and 0.050 across context-window sizes. For consistency with the primary configuration selected by the quality–balance analysis (Section 4.2), the core experiments use $p = 1.0$, $q = 0.5$, $d = 2$ with $L = 10$, $\gamma = 80$, and $w = 5$. The sensitivity analysis nevertheless indicates that a larger walk length (particularly $L = 80$) can provide further improvement for this dataset.

4.7 Ablation: Alternative Node2Vec Configurations

To complete the configuration analysis of Section 4.2, we examine the two alternatives to the primary setting.

Quality-optimal configuration ($p = 2.0$, $q = 1.0$, $d = 1$).

If clustering quality is the sole criterion, this configuration is the clear winner: it achieves per-course Silhouette Scores of 0.963, 0.593, 0.927, 0.634, 0.838, and 0.613 (average 0.765 at seed 42; 0.755 ± 0.037 over 20 seeds). Its sections, however, are impractical without an explicit balancing constraint: courses 1 and 3 collapse into splits of 72/2 and 47/1 students, the average balance over 20 seeds is only 0.406, and the per-course average balance drops to 0.021 for course 3. Its clustering stability is also the lowest of the three candidates (average KMeans ARI 0.980), driven by the genuinely bimodal structure of course 6 at $d = 1$, where GMM exhibits pronounced variability (ARI 0.442) between the two competing one-dimensional partitions. These results confirm that the high Silhouette Scores at $d = 1$ partly reflect near-singleton clusters rather than practically usable sections.

Neutral configuration ($p = 1.0$, $q = 1.0$, $d = 2$).

The DeepWalk-equivalent configuration is a natural reference point. It achieves an average Silhouette Score of 0.629 at seed 42 (0.618 ± 0.012 over 20 seeds) with an average balance of 0.558 (seed 42) and stable clusterings (average ARI 0.984). As

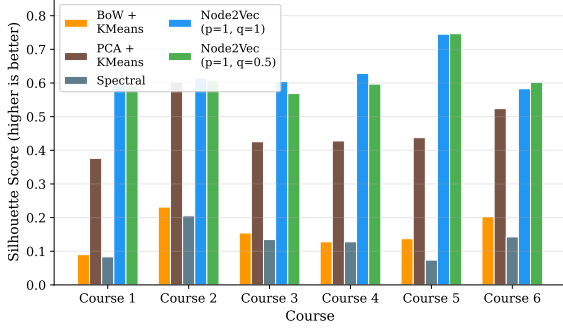


Fig. 7: Silhouette score ($\uparrow$ higher is better) across the six courses for the conventional Student-Course representation (BoW + KMeans), PCA + KMeans, spectral clustering, and Node2Vec under the neutral ($p=1, q=1$) and primary ($p=1, q=0.5$) configurations, all with $k=2$ KMeans clustering.

Table 2 shows, it is dominated by the primary configuration, which provides a higher Silhouette over repeated seeds, a higher worst-course balance, and higher stability at an essentially identical average balance.

The ablation confirms the selection made in Section 4.2: the quality-optimal configuration demonstrates the representational power of the co-enrollment graph but yields unusable section sizes, whereas the primary configuration retains a substantial advantage over all baselines in both clustering quality and section balance with the highest stability.

4.8 Clustering Algorithms and Representation Quality

To determine whether the observed improvement depends on a particular clustering algorithm, we additionally evaluate the learned Node2Vec representations using different clustering methods; the corresponding stability analysis is reported in Table 11. Figure 7 summarizes the per-course Silhouette scores obtained when each student representation—the conventional Student-Course representation (BoW), its PCA projection, spectral clustering applied directly to the co-enrollment graph, and the Node2Vec embeddings under the neutral ($p=1, q=1$) and primary ($p=1, q=0.5$) configurations—is combined with KMeans clustering.

Node2Vec consistently produces higher clustering quality than the conventional representation across the six courses, and the primary configuration ($p=1.0, q=0.5$) also outperforms PCA+KMeans on every course (Section 4.10). The advantage is therefore associated with the learned student representation rather than with the particular downstream clustering stage; the robustness of the resulting clusterings across clustering algorithms is examined separately through the stability analysis in Table 11.

This result is consistent with the formulation of the proposed framework, in which graph representation learning is used as a preprocessing stage and the downstream clustering procedure is deliberately kept unchanged. The purpose is therefore not to introduce a specialized clustering algorithm, but to provide a representation in which students with structurally similar enrollment patterns become more suitable for subsequent grouping.

Combined with the PCA comparison reported in Table 3, these results provide a more complete interpretation of the role of representation quality. PCA substantially improves clustering relative to the original high-dimensional enrollment representation, showing that dimensionality reduction is beneficial. Nevertheless, Node2Vec achieves a higher average Silhouette Score than PCA+KMeans. The additional improvement suggests that the graph-based embedding captures relational information that is not fully recovered by applying a generic linear dimensionality-reduction technique to the enrollment matrix.

4.9 Clustering Stability and Additional Evaluation Metrics

In addition to clustering quality, reproducibility is important in student sectioning because unstable assignments may lead to inconsistent student groups across repeated runs. We therefore evaluate clustering stability across 20 random initializations using the Adjusted Rand Index (ARI): the Node2Vec embeddings are computed once with the primary configuration ($p=1.0, q=0.5, d=2$) at the reference seed, and the clustering stage is re-run under 20 random seeds. The results are presented in Table 11.

The Node2Vec representation produces highly stable clusterings across repeated runs. Under the

Table 11: Clustering stability measured by Adjusted Rand Index (ARI) across 20 random initializations ($\uparrow$ higher is better). Node2Vec columns use the primary configuration ($p = 1.0$, $q = 0.5$, $d = 2$); BoW applies KMeans directly to the raw enrollment representation.

Course	Node2Vec , ARI			BoW
	KMeans	GMM	Agglom.	KMeans
1	1.000	1.000	1.000	0.381
2	1.000	0.738	1.000	1.000
3	1.000	0.904	1.000	0.852
4	1.000	0.738	1.000	0.985
5	0.961	1.000	1.000	0.860
6	1.000	1.000	1.000	1.000
Avg.	0.994	0.897	1.000	0.846

primary configuration ($p = 1.0$, $q = 0.5$, $d = 2$), KMeans achieves an average ARI of 0.994, with the lowest per-course value of 0.961 on Course 5, while agglomerative clustering, which is deterministic given the embeddings, achieves an ARI of 1.000 across the six courses. GMM attains an average ARI of 0.897, with slightly greater variability on Courses 2 and 4 (0.738 each). In comparison, KMeans applied to the conventional Student-Course Representation has a lower average ARI of 0.846, with particularly low stability for Course 1 (0.381). For reference, the quality-optimal configuration analyzed in Section 4.7 achieves a lower average KMeans ARI of 0.980, with pronounced GMM instability on Course 6, reflecting the bimodal structure of that course at $d = 1$.

We further evaluate clustering quality using the Davies–Bouldin Index (DBI) and the Calinski–Harabasz (CH) Index. Figures 8 and 9 present the corresponding comparisons.

The DBI and CH results provide complementary evidence regarding the structure of the obtained clusters. Across the evaluated courses and clustering methods, Node2Vec generally produces more favorable DBI and CH values than the conventional representation. These results are consistent with the Silhouette analysis and indicate that the observed improvement is reflected in multiple internal clustering criteria rather than in a single evaluation metric.

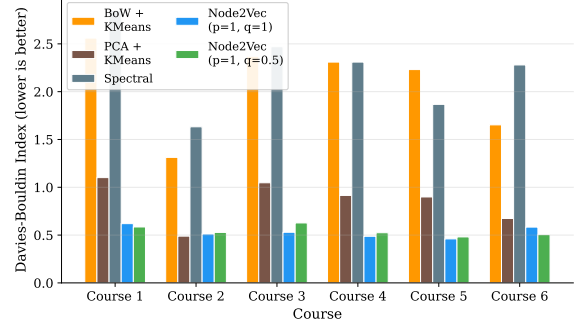


Fig. 8: Davies–Bouldin Index ($\downarrow$ lower is better) across the six courses for different clustering methods and representations.

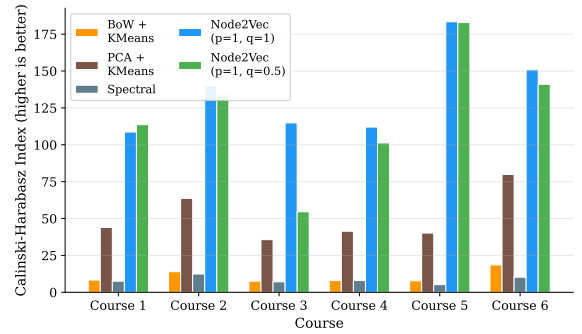


Fig. 9: Calinski–Harabasz Index ($\uparrow$ higher is better) across the six courses for different clustering methods and representations.

4.10 Practical Significance and Consistency

Due to the inherently limited number of course-specific sectioning instances available from a single institutional dataset ($n = 6$), we focus on effect sizes and consistency of improvement across courses rather than formal statistical hypothesis testing. This approach is appropriate for educational data mining, where the number of large courses requiring complex sectioning in a single institution is naturally limited.

Node2Vec versus conventional representation.

Node2Vec (primary configuration $p=1.0$, $q=0.5$, $d=2$) outperforms the conventional BoW representation on all six courses, with Silhouette Scores substantially higher on every evaluated course.

This consistency provides strong evidence for practical significance.

Node2Vec versus PCA+KMeans.

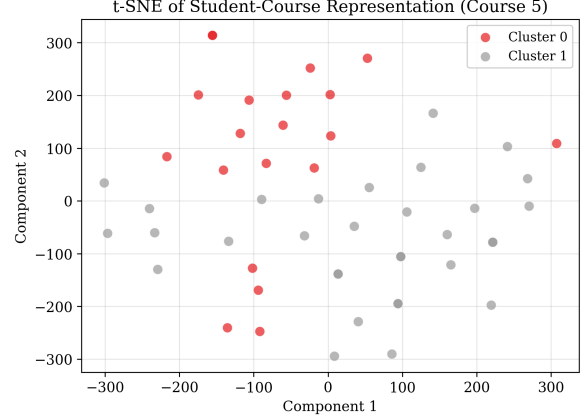
Node2Vec (primary configuration) also outperforms PCA+KMeans on all six courses at the reference seed, with an average relative improvement of 33% in Silhouette Score. The margin is large on most courses; on Course 2 the two methods are nearly tied (0.608 vs. 0.603), whereas the remaining courses show clear gaps. The consistent average advantage therefore reflects a genuine benefit of the graph-based representation over linear dimensionality reduction, while the Course 2 near-tie should not be overstated.

In summary, combined with the large average magnitude of improvement, the per-course consistency provides strong evidence for practical significance. In educational scheduling applications, consistent improvements in representation quality can support better student groupings for subsequent timetable construction, although end-to-end timetable quality is not evaluated in this study.

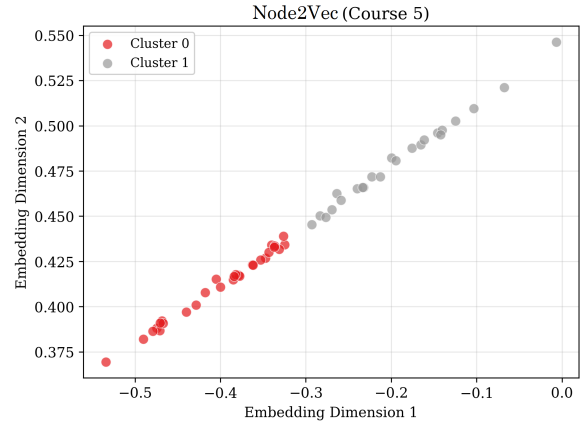
4.11 Embedding Visualization

To provide a qualitative interpretation of the quantitative results, we visualize the learned representations for a sample course. The visualizations compare the conventional Student-Course representation with the two-dimensional Node2Vec representation. For the conventional Student-Course representation, which is defined in a higher-dimensional space, t-Distributed Stochastic Neighbor Embedding (t-SNE) is used solely for visualization. In contrast, Node2Vec directly produces a low-dimensional embedding, and the primary configuration ($d = 2$) is itself two-dimensional, so no additional dimensionality-reduction step is required. The embedding shown in Figure 10 uses the neutral configuration ($p = 1.0$, $q = 1.0$, $d = 2$); the primary configuration differs only in the walk bias and admits the same direct 2D visualization.

Figure 10 presents the representations obtained for Course 5. The visualization of the conventional representation shows a relatively overlapped and dispersed structure, whereas the Node2Vec representation exhibits more compact and visually separated groups. This qualitative



(a) t-SNE of the *Student-Course Rep.*



(b) Node2Vec embeddings ($p=1.0$, $q=1.0$, $d=2$; neutral configuration).

Fig. 10: Visualization of student representations for Course 5. (a) The conventional Student-Course representation is defined in a 32-dimensional space and is projected to two dimensions using t-SNE solely for visualization. (b) Node2Vec directly produces a two-dimensional representation, so no additional dimensionality-reduction method is required. In both panels, students are displayed according to their resulting section assignments.

difference is consistent with the substantially higher Silhouette Score obtained by Node2Vec for this course.

These visualizations provide qualitative support for the quantitative results, while the distinction between the two visualization procedures should be noted: t-SNE is applied only to the high-dimensional conventional representation, whereas

the Node2Vec points are plotted directly in their learned two-dimensional embedding space, without any additional projection.

4.12 Runtime Analysis

Finally, we examine the computational cost of the complete Node2Vec pipeline. Table 12 reports the runtime of graph construction, random-walk generation, Word2Vec training, and the complete pipeline for each course. The complete pipeline requires less than nine seconds for all six courses. Word2Vec training is the dominant component of the runtime, accounting for approximately 96.5% of the total execution time, whereas graph construction and walk generation require comparatively little time.

For a dataset with n students and m courses, construction of the student co-enrollment graph from the enrollment matrix requires comparing all student pairs over all courses, yielding a cost of $O(n^2m)$. Random-walk generation requires $O(n\gamma L)$ operations, where γ is the number of walks per node and L is the walk length. The cost of Skip-Gram training is proportional to the total number of walk-context pairs and the embedding dimension, approximately $O(n\gamma Ld)$, where d is the embedding dimension. For the relatively small to moderate student populations considered in this study, the overall computational cost remains modest.

Overall, the experiments demonstrate that the Node2Vec representation provides substantially improved clustering quality relative to the conventional Student-Course representation in the evaluated setting. The improvement is observed across multiple clustering algorithms and evaluation metrics, while the stability analysis indicates that the resulting clusterings are highly reproducible. The sensitivity experiments further show that effective representations can be obtained in a very low-dimensional space. The runtime analysis indicates that the complete pipeline can be executed within a few seconds for the course sizes considered in this study.

5 Discussion

The experimental results provide consistent evidence that graph representation learning substantially improves the quality of student sectioning.

Table 12: Runtime analysis in seconds for the complete Node2Vec pipeline using $d = 2$, $L = 10$, $\gamma = 80$, $w = 5$, and $\epsilon = 30$.

Course	Students	Graph	Walks	Word2Vec	Total
1	74	0.024	0.263	8.067	8.355
2	51	0.011	0.168	5.471	5.650
3	48	0.009	0.144	4.584	4.738
4	49	0.070	0.145	4.785	5.000
5	52	0.009	0.153	4.830	4.992
6	67	0.016	0.218	6.564	6.798

Across all evaluated courses and clustering algorithms, Node2Vec produces markedly higher clustering quality than both the conventional student-course representation and spectral clustering.

A key explanation for this improvement lies in how relational information is captured. The conventional representation describes students only through direct course enrollments, representing relationships implicitly. In contrast, the co-enrollment graph explicitly models students as nodes connected by weighted edges. Random walks on this graph expose the embedding model to sequences of structurally related students, allowing Node2Vec to encode higher-order proximity and contextual relationships not expressed in the original enrollment matrix.

The baseline experiments provide an important control. While PCA substantially improves clustering over the conventional representation by reducing dimensionality, Node2Vec achieves consistently higher scores even when projected to the same two-dimensional space. This indicates that the performance gain is not merely a byproduct of dimensionality reduction, but rather stems from the structural graph-context captured by the random-walk-based learning. Similarly, the poor performance of direct spectral clustering confirms that simply applying a graph partitioning algorithm to the co-enrollment graph is insufficient; learning a task-oriented representation is critical.

An important finding is that the best performance is achieved with very low-dimensional embeddings, particularly $d = 1$ and $d = 2$. This suggests that the structural patterns relevant to

student sectioning in this dataset are highly compressible. The quality–balance Pareto analysis, however, shows that the two objectives do not share the same optimum: configurations at $d = 1$ maximize the Silhouette Score (up to 0.765) but produce severely imbalanced sections, whereas the primary configuration adopted throughout this paper ($p = 1.0$, $q = 0.5$, $d = 2$) provides the best trade-off, combining the highest multi-seed Silhouette among balanced candidates with more balanced sections than all baselines and the highest clustering stability.

The stability of the learned representations across different clustering algorithms (Table 11) indicates that Node2Vec serves as an effective pre-processing stage compatible with standard institutional clustering pipelines. Furthermore, the high stability across random seeds and the modest computational cost (under 9 seconds per course) support the practicality of the framework as an offline representation stage for further evaluation in educational scheduling workflows.

6 Conclusion

This paper revisited the student sectioning problem from a graph representation learning perspective. By constructing a weighted student co-enrollment graph and employing Node2Vec, we learned low-dimensional student embeddings that substantially improve clustering quality compared to conventional representations and spectral clustering.

Experiments on six real-world university courses demonstrated that Node2Vec outperforms the strongest baselines, including PCA followed by KMeans, with a large average advantage and wins on all six courses at the reference seed. The proposed framework shows that the structural information captured by biased random walks provides additional value beyond generic dimensionality reduction. The method exhibits stable performance across the tested random-walk parameters, with strong dependence on embedding dimension, high stability across random seeds (average KMeans ARI of 0.994 under the primary configuration), very low computational cost, and a quality–balance configuration analysis that identifies a setting providing both strong clustering quality and practically balanced section sizes.

The framework can be seamlessly incorporated into existing timetabling pipelines without modifying downstream clustering algorithms. Future work will explore the application of this framework to larger, multi-institutional datasets, sparser co-enrollment graphs, and the integration of graph neural networks to incorporate additional student attributes. Ultimately, integrating graph-based sectioning with constraint programming could allow direct evaluation of its impact on complete timetable quality.

7 Limitations

Several limitations of the present study should be acknowledged. First, the experimental evaluation relies on clustering-based internal validity measures (Silhouette Score, DBI, and CH Index), which assess the coherence and separation of the resulting groups. While these metrics are appropriate for evaluating the quality of student representations and groupings, they do not directly capture all practical aspects of student sectioning such as section balance, capacity constraints, student-course conflicts, or downstream timetable feasibility. To partially address this, we compute a simple section balance metric $B = \min(|S_1|, |S_2|) / \max(|S_1|, |S_2|)$ for the two resulting sections. For the primary configuration, the computed values range from 0.368 to 0.815 (average 0.652) across the six courses, exceeding all baselines (average B of 0.465–0.471); the quality-optimal ablation configuration, in contrast, ranges from 0.021 to 0.675 (average 0.308). While this metric provides a useful characterization of practical suitability, a comprehensive end-to-end evaluation of sectioning quality within a complete timetabling pipeline remains future work.

Second, the study uses a single institutional dataset comprising 210 students and 38 courses, from which six course-specific sectioning instances are extracted. While this dataset provides a realistic testbed, the limited number of instances and the shared student population mean that statistical independence across instances cannot be assumed. Future work should evaluate the proposed approach on larger, multi-institutional datasets with diverse enrollment patterns.

Third, the clustering stage is deliberately kept as standard KMeans (or other conventional algorithms) to isolate the effect of the representation.

In practice, more sophisticated sectioning algorithms that incorporate institutional constraints could be combined with the learned representations to achieve further improvements.

Finally, the primary configuration was selected through a quality–balance Pareto analysis on the same six courses used for performance evaluation, while the silhouette-optimal configuration ($d = 1$, $p = 2.0$, $q = 1.0$) is reported as an ablation. Configuration selection is therefore in-sample on this dataset, and the appropriate weighting of clustering quality and section balance may be application-specific.

8 Declarations

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the author utilized generative AI technologies for various aspects of the research and manuscript preparation process, including writing and editing assistance, code development and language editing. All content and analyses were reviewed, verified, and finalized by the author, who assumes full responsibility for the manuscript

Competing interests

The author has no relevant financial or non-financial interests to disclose.

Funding

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Ethics approval

Not applicable. The study uses historical, de-identified course enrollment records for retrospective computational analysis.

Consent to participate

Not applicable. No new human participants were recruited for this study.

Consent for publication

Not applicable.

Code and Data availability

The Python code and data used in this work are available at <https://github.com/mamintoosi/Node2Vec-SSP>.

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A Node2Vec Random Walk Generation

Algorithm 2 details the generation of biased random walks used in this study.

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Algorithm 2 GenerateRandomWalks (Node2Vec)

Require: Graph $G = (V, E, W)$, walk length L , number of walks per node γ , return parameter p , in-out parameter q

Ensure: A list $\mathcal{W}$ containing all sequences of random walks

```
1:  $\mathcal{W} \leftarrow []$  ▷ Initialize list to store walks
2: for each graph node as start do
3:   for  $k = 1$  to  $\gamma$  do ▷ Number of walks per node
4:      $walk \leftarrow [start]$  ▷ Start the walk at the current node
5:      $prev \leftarrow \text{null}, cur \leftarrow start$ 
6:     for  $j = 1$  to  $L$  do ▷ Walk length
7:        $N(cur) \leftarrow \text{neighbors of } cur \text{ in } G$ 
8:       if  $prev = \text{null}$  then ▷ First step: proportional to edge weight
9:         Sample  $next$  from  $N(cur)$  with probability  $\propto w_{cur,next}$ 
10:      else ▷ Compute biased transition probabilities
11:
12:        for each neighbor  $z \in N(cur)$  do
13:           $d \leftarrow \text{shortest-path distance from } z \text{ to } prev \text{ in } G$ 
14:           $\alpha_{pq} \leftarrow \begin{cases} 1/p, & d = 0 \\ 1, & d = 1 \\ 1/q, & d = 2 \end{cases}$ 
15:           $\pi(z) \leftarrow \alpha_{pq} \cdot w_{cur,z}$  ▷ Unnormalized weight
16:        end for
17:        Normalize  $\pi$  into a probability distribution over  $N(cur)$ 
18:         $next \leftarrow \text{sample from } N(cur) \text{ according to } \pi$ 
19:      end if
20:       $walk \leftarrow walk + [next]$  ▷ Append next node to walk
21:       $prev \leftarrow cur, cur \leftarrow next$ 
22:    end for
23:     $\mathcal{W} \leftarrow \mathcal{W} + [walk]$  ▷ Store the walk
24:  end for
25: end for
26: return  $\mathcal{W}$ 
```

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